SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I Patteela Yashwanth Kumar/192471055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Patteela Yashwanth kumar*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Compare L1, L2 and L3 Cache in Terms of Latency, Bandwidth and Throughput

Introduction

Cache memory is a high-speed memory located close to the processor. Modern processors use multiple cache levels—**L1, L2 and L3**—to reduce the time required to access frequently used data and instructions.

Comparison

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside CPU core	Near/inside core	Usually shared by cores
Capacity	Small	Medium	Large
Latency	Very low	Low	Higher than L1/L2
Bandwidth	Very high	High	Moderate to high
Throughput	Highest	High	Lower than L1/L2
Cost/bit	Highest	High	Lower
Main purpose	Immediate CPU access	Backup for L1	Backup for L1 and L2

L1 Cache

L1 is the **smallest and fastest** cache. It is usually divided into instruction cache and data cache. Its very low latency allows the processor to access frequently required data quickly.

L2 Cache

L2 is larger than L1 but slightly slower. When data is not found in L1, the processor searches L2. It provides a balance between **capacity and access speed**.

L3 Cache

L3 is generally the largest cache and is often shared among multiple processor cores. It has higher latency than L1 and L2 but is still faster than accessing DRAM.

Impact on Processor Performance

A cache hit allows the processor to obtain data quickly, increasing throughput. A cache miss causes the processor to access the next cache level or DRAM, increasing waiting time.

Conclusion

L1, L2 and L3 work together as a hierarchy. Increasing cache hit rate reduces memory access latency and improves overall processor performance and throughput.

2. Differentiate SRAM and DRAM

Introduction
SRAM and DRAM are two important types of semiconductor memory. **SRAM is mainly used for cache memory**, while **DRAM is mainly used as main memory**.

Comparison

Feature	SRAM	DRAM
Storage	Flip-flop circuit	Capacitor
Latency	Very low	Higher
Bandwidth	Very high	High
Throughput	Very high	High
Refresh	Not required	Required
Density	Lower	Higher
Cost	Expensive	Less expensive
Application	Cache	Main memory

SRAM
SRAM stores each bit using a flip-flop circuit. It does not require periodic refreshing. Therefore, it provides **very low latency and high throughput**.
Its major disadvantage is that it requires more transistors per bit, making it expensive and less dense. Hence, SRAM is suitable for relatively small cache memories.

DRAM
DRAM stores each bit using a capacitor and transistor. The capacitor loses charge over time, so DRAM requires periodic refreshing. This increases access overhead.
However, DRAM has **high density and lower cost per bit**, making it suitable for large main-memory systems.

Performance Analysis
SRAM is faster and provides higher bandwidth, making it suitable when the processor requires immediate access. DRAM sacrifices some speed to provide much larger capacity.
For example:
CPU → SRAM Cache → DRAM Main Memory
This arrangement reduces the number of slow DRAM accesses.

Conclusion
SRAM is preferred for **low latency and high-speed cache operation**, whereas DRAM is preferred for **large capacity and economical main memory**

3. Compare Virtual Memory with Paging and Cache Memory

Introduction

Virtual memory and cache memory both improve memory management and performance, but they serve different purposes. **Paging manages virtual memory**, while **cache memory reduces CPU memory-access latency**.

Comparison

Feature	Virtual Memory/Paging	Cache Memory
Main purpose	Provides larger logical memory	Reduces access latency
Managed by	OS + MMU	Hardware
Storage	DRAM + secondary storage	SRAM
Access unit	Page	Cache block/line
Latency	High during page fault	Very low on cache hit
Capacity	Very large	Small
Throughput	Reduced during page faults	Increased by cache hits
Main benefit	Capacity and protection	Speed

Virtual Memory and Paging

Virtual memory allows a program to use an address space larger than the available physical RAM. Paging divides virtual memory into fixed-size **pages** and physical memory into **frames**. When a required page is absent from RAM, a **page fault** occurs. The operating system loads the page from secondary storage.

Cache Memory

Cache memory stores frequently accessed instructions and data. If the required data is present, a **cache hit** occurs and the processor gets the data quickly.
If it is absent, a **cache miss** occurs and the next memory level must be accessed.

Critical Analysis

Cache memory primarily improves **access latency and processor throughput**. Virtual memory primarily provides **memory capacity, isolation and protection**.
A cache miss usually causes a relatively small delay compared with a page fault. A page fault can cause a very large delay because secondary storage is much slower than DRAM.

Conclusion

Cache and virtual memory complement each other:
CPU → Cache → DRAM → Virtual Memory/Secondary Storage
Cache improves speed, while paging provides efficient memory management and a larger logical address space

4. Analyze NAND Flash and DRAM

Introduction
NAND Flash and DRAM are different memory technologies used at different levels of the memory hierarchy. DRAM is a **volatile working memory**, whereas NAND Flash is a **non-volatile storage technology**.

Comparison

Feature	DRAM	NAND Flash
Type	Volatile	Non-volatile
Latency	Very low	Higher
Bandwidth	High	High, especially sequential
Throughput	High	Moderate to high
Write operation	Fast	Slower
Data retention	Requires power	Retains data without power

DRAM
DRAM provides fast read and write access and is directly used as the system's main memory. It offers low latency and high throughput, allowing the CPU to execute programs efficiently. However, DRAM is volatile and loses its contents when power is removed.

NAND Flash
NAND Flash stores data even when power is removed. It provides high storage density and is widely used in SSDs.
Its major limitation is higher access latency compared with DRAM, especially for random operations. Flash also has limited write endurance.
memory system, DRAM is placed closer to the processor and NAND Flash is used at the lower storage level.

Hierarchical Suitability
A typical memory hierarchy is:
CPU → Cache → DRAM → NAND Flash
DRAM is placed closer to the processor because it requires faster access. NAND Flash is placed lower in the hierarchy because it provides much larger persistent storage.

Comparative Analysis – NAND Flash vs DRAM

NAND Flash and DRAM differ mainly in **latency, bandwidth, throughput, and data retention**. DRAM is volatile and provides very low latency, high bandwidth and high throughput, making it suitable for main memory. NAND Flash is non-volatile, has higher latency and slower write operations, but provides high sequential bandwidth and moderate-to-high throughput. DRAM requires continuous power to retain data, whereas NAND Flash retains data even without power. Therefore, DRAM is preferred for **fast working memory**, while NAND Flash is suitable for **large-capacity permanent storage**. In a hierarchical memory system, DRAM is placed closer to the processor and NAND Flash is used at the lower storage level.

Critical Analysis

DRAM provides **very low latency, high bandwidth, and high throughput**, making it suitable for frequently accessed working data. However, it is volatile and requires continuous power. NAND Flash has **higher latency and slower write operations**, but it is non-volatile and retains data without power. Therefore, using DRAM for speed and NAND Flash for persistent, high-capacity storage provides a better balance of **performance, capacity, and data retention** in a hierarchical memory system.

Conclusion

DRAM and NAND Flash are complementary technologies. DRAM improves system performance as main memory, while NAND Flash provides high-capacity permanent storage. DRAM is best suited for fast main memory because of its very low latency and high throughput, while NAND Flash is suitable for large, non-volatile storage due to its higher capacity and data retention.

Both technologies complement each other in a hierarchical memory system.In conclusion, DRAM is suitable for fast main-memory access due to its low latency and high throughput, while NAND Flash is better for large, non-volatile storage. Both technologies complement each other to provide an efficient hierarchical memory system.

5. Compare MRAM and RRAM

Introduction

MRAM and RRAM are emerging **non-volatile memory technologies** designed to provide faster and more energy-efficient alternatives to conventional memory and storage technologies.

Comparison

Feature	MRAM	RRAM
Storage principle	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access time	Very fast	Fast
Read throughput	High	High
Write throughput	High	Moderate to high
Energy efficiency	High	Potentially very high
Endurance	Very high	High, technology-dependent
Scalability	Good	Very good
Applications	Embedded memory, cache	Embedded memory, cache

MRAM

- MRAM stores information using magnetic states. Since it is non-volatile, it retains information even when power is removed.
- Its important advantages are:
- Fast access
- High endurance
- Low standby power
- Non-volatility
- Good reliability

RRAM

- stores information by changing the electrical resistance of a material. It can provide high density and low-power operation.
- Its advantages include:
- Fast switching
- High density
- Low energy requirement RRAM

- Good scalability
- Non-volatile operation

Critical Analysis
MRAM is particularly strong in **speed, endurance and reliability**. RRAM has strong potential in high-density and energy-efficient memory systems. However, RRAM performance and endurance depend on the specific implementation.MRAM offers **very fast access, high read/write throughput, high energy efficiency, and very high endurance**, making it suitable for performance-critical memory application.

RRAM provides **fast access, high read throughput, potentially very high energy** efficiency, and very good scalability, but its endurance is technology-dependent. Therefore, **MRAM is preferable when speed and** endurance are priorities, while RRAM is promising for scalable and energy-efficient memory systems.

Conclusion
MRAM provides very fast access, high throughput, and very high endurance, while RRAM offers fast access, good scalability, and potentially very high energy efficiency. Both are promising non-volatile memory technologies for **future high-performance and energy-efficient memory systems.**
MRAM and RRAM can help bridge the gap between volatile memory and persistent storage. MRAM emphasizes speed and endurance, while RRAM offers strong potential for density, scalability and energy efficiency.